Felipe B. Castro

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FULL-STACK DEVELOPER | LLMs | DATA | DEVOPS | LOW-LEVEL

Summary

Software Engineering student (7th semester, IDP) who likes to understand how things work. Full-stack developer and software architect working across web applications, LLMs and AI agents, data pipelines, DevOps and cloud, and low-level systems. Owns systems end to end, from architecture decisions to production, and turns raw data into decision-making tools. Hackathon and CTF enthusiast with several podium finishes. English advanced.

Experience

Nex AI 08/2026 - Present
Software Architect, Tech Lead & Full-Stack Developer *Remote*

- Lead software architecture and technical direction of an AI product, from design to delivery, with multi-agent LLM orchestration and human-in-the-loop review gates.
- Authored 20+ Architecture Decision Records reviewed with the Product Owner, covering DAG-based parallel task scheduling and convergence, durable commands and events between services, immutable intents with idempotent retries, hybrid persistence with projections, and polling with snapshots.
- Designed an AWS edge with CloudFront as the single entry point, a static frontend on S3, and API traffic routed to an ALB behind WAF.
- Technical reviewer of the AI layer: document chunking and embeddings, per-agent model tiers, persistent agent-loop state, and token and cost observability.

Nexara Tech 07/2026 - 09/2026
Web Developer, Branding & Digital Marketing *Brasília*

- Built the Nexara/NexAI presentation website and an interactive product mockup: Next.js 16, React 19, Tailwind v4, Motion, TypeScript, i18n in three languages (EN/PT/ES).
- Prototyped an interactive workflow flowchart in React/TypeScript and wrote commercial proposals translating technical architecture into business narratives.
- Paid traffic and organic growth for a political campaign, with AI-assisted content generation, data analysis and automation of posts and processes.

Ministry of Education (MEC) 06/2026 - Present
Full-Stack Developer *Brasília*

- Sole developer at the Internal Affairs Office, owning the full lifecycle from requirements to deployment.
- Built systems that turn raw case data into strategic KPIs, including a document redaction system and a predictive jurimetrics module.
- Data governance: data glossary and BPMN 2.0 operational workflows.

JusBrasil (IDP Sprint) 1st Place in Innovation 04/2026 - 06/2026
Tech Lead & Full-Stack Developer *Brasília*

- Clean Architecture tool with an asynchronous, distributed pipeline that classifies and normalizes legal research outputs, processing thousands of records.
- Cut the time JusBrasil spent on the task by 99.8% (about 323% monthly ROI) in 12 weeks.
- Stack: FastAPI, PostgreSQL, SQLAlchemy, Docker, AWS S3, JWT, Next.js.

Ize Gestão Empresarial 10/2025 - 12/2025
Data Intern *Brasília*

- Econometric indicators for financial consultancies; ETL with Airflow, scraping with Selenium, PostgreSQL, Docker.

Oficina Consultorias 05/2025 - 10/2025
IT Support Intern *Brasília*

- Network, server and corporate system administration (Azure, Docker, Python).

Projects

Linux Kernel Syscall & Char Driver GitHub
C, Linux 6.6, QEMU, BusyBox

- Patched and built a Linux 6.6 kernel with a new syscall that interacts with a character driver and elevates the calling process to root; booted on a custom BusyBox initramfs under QEMU (kernel lab).

x86 Bootloader GitHub
x86 Assembly, real mode, QEMU, GDB

- 512-byte real-mode bootloader that loads a kernel at `0x7E00`, passes state through `AX` and jumps to it; debugged via the QEMU GDB stub.

Keep Solving and Nobody Explodes GitHub

C, POSIX threads, ncurses, SDL2

- Concurrent bomb-defusal game where timer, modules, UI and audio run as independent POSIX threads synchronized over shared state.

Converge GitHub

Rust, Tauri 2, TypeScript

- Local-first desktop app to monitor multi-agent coding workflows: native Rust core with transport-independent domain, Rust parsers for agent session logs, and a capability-scoped IPC boundary (no generic shell or filesystem access for the UI).

Second Brain Private repository

Obsidian, Claude Code, MCP, Git

- AI-augmented knowledge base in Obsidian with 30+ plugins (Dataview, Templater, Tasks, Excalidraw, Obsidian Git), a flat tag system, Maps of Content and a path-based color scheme for the Graph View.
- Built 9 custom Claude Code skills (capture and refile, link and tag audits, Notion import, session distillation, Git sync with union-merge of conflicts) plus session hooks, connected to the vault through an MCP server over the Local REST API plugin.
- Token-cost discipline: identified context volume as about 73% of AI spend and cut it with search-first reads and patch-based edits; AI-spend dashboard generated from usage logs.

SQLMap Seminar GitHub

SQL injection, Docker, DVWA

- Study material and live demo of automated SQL injection detection and exploitation against an isolated DVWA lab in Docker.

Competitions

Ekoparty CTF 2026, Main CTF (Buenos Aires) 4th Place Writeups

Team `nop` (2 players), Web3 specialist: 11 solves, 3320 pts, 2 first bloods *10/2026*

- EVM exploits: price-oracle manipulation, flash-loan governance attack, reentrancy, chained proxy/`delegatecall` bugs and signed-voucher replay; also worked on pwn (memory over-read) and malware reversing with Ghidra.
- **CTF Hackathon, 6th Edition** (10/2025): Pwn, Rev, Crypto, Jail, Web, Net, Stego.
- **Brasília+IT Robotics, Factory Floor, 1st Place** (08/2025).
- **IDP Hackathon, 5th Edition, 1st Place** (06/2025): AI platform against restaurant food waste (Flask, Ollama, MongoDB).
- **Campus Party Hackathon** (08/2025): 72h build, incident heat map for women's safety.
- **IDP GO Silicon Valley** (07/2024): selected for program with Google, Meta, Microsoft, Uber.

Education

Instituto de Desenvolvimento e Pesquisa (IDP) 08/2023 - 08/2027 (expected)

B.Eng. Software Engineering, 7th semester *Brasília*

- Relevant coursework: Operating Systems, Concurrent Programming, Algorithm Analysis, Software Security & Testing.

Skills

- **Languages:** Python, TypeScript, JavaScript, C, C++, x86 Assembly, Rust (learning), Java, C#, SQL
- **Frontend:** Next.js, React, Angular, Tailwind, Motion
- **Backend:** FastAPI, Flask, NestJS, Node.js, Pydantic, REST APIs, JWT, Clean Architecture, event-driven design
- **Data:** PostgreSQL, SQLAlchemy, MongoDB, Redis, Cassandra, Supabase, Airflow, Power BI, Selenium
- **DevOps & Cloud:** Docker, Linux, AWS (CloudFront, WAF, ALB, S3), Azure, Kafka, RabbitMQ, MinIO, Git
- **LLMs & AI:** multi-agent orchestration, RAG, embeddings, Ollama, PyTorch, scikit-learn, n8n
- **AI Tooling:** Claude Code (skills, hooks, subagents), MCP servers, OpenAI Codex, Spec Kit (Spec-Driven Development)
- **Low-Level:** Linux kernel (syscalls, modules), QEMU, POSIX threads, GDB/pwndbg, pwntools, checksec, objdump, strace/ltrace
- **Security & Reverse:** Ghidra, radare2, binwalk, z3, Burp Suite, nmap, Wireshark, CyberChef, hashcat, John the Ripper, SQLMap
- **Web3:** Solidity, Foundry (forge, cast, anvil), web3.py
- **Build & Docs:** Bash, Make, Just, Vite, Tauri, TanStack Query, C4 Model, BPMN 2.0, Mermaid, PlantUML, LaTeX
- **QA:** Pytest, Playwright, SonarQube, Gherkin
- **Spoken:** Portuguese (native), English (advanced), Spanish (basic)